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import tensorflow as tf
from tensorflow.keras import layers, models
import matplotlib.pyplot as plt

# Load the MNIST dataset
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()

# Normalize the data
x_train = x_train / 255.0
x_test = x_test / 255.0

# Reshape data to fit the model: (28, 28, 1)
x_train = x_train.reshape((-1, 28, 28, 1))
x_test = x_test.reshape((-1, 28, 28, 1))

# Build the CNN model
model = models.Sequential([
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(28, 28, 1)),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

# Compile the model
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# Train the model
model.fit(x_train, y_train, epochs=5, validation_data=(x_test, y_test))

# Evaluate the model
test_loss, test_acc = model.evaluate(x_test, y_test)
print(f"Test accuracy: {test_acc:.4f}")

# Make predictions on the test set
predictions = model.predict(x_test)

# Show some predictions
for i in range(5):
    plt.imshow(x_test[i].reshape(28,28), cmap='gray')
    plt.title(f"Predicted: {tf.argmax(predictions[i]).numpy()}, Actual: {y_test[i]}")
    plt.axis('off')
    plt.show()

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